FIDELITYLOOP: CLOSED-LOOP VALIDATION OF ELASTIC LLM-SERVING POLICIES

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ABSTRACT

Elastic LLM-serving policies change the capacity on which their own queues and control decisions depend. Accurate request-time predictions therefore do not establish whether a policy will meet deadlines after deployment. We present FidelityLoop, a closed-loop validation framework that runs rules, a deadline guard, and replay-trained PPO policies through a shared interface. Request identities align service timing, resource trajectories, control actions, and deadline-aware costs across replay, vLLM, and SGLang execution. In 36 preregistered runs on two A100 GPUs, a calibrated replay model improves matched-request timing while worsening occupied-resource prediction. Startup–cooldown interactions explain altered actions; retrospective replay with a Vidur service backend also exposes timing errors hidden by aggregate agreement. Tests on frozen PPO checkpoints, additional arrivals, and a 14B port show that a startup proxy that detects failed policies in one context can reject feasible policies or accept failing ones in another. These errors have practical consequences: retrospective replay selection chooses a policy that fails all six physical repeats; in another tested pool, it selects a feasible policy costing 58.95% more than the lowest-cost feasible alternative under the scenario ledger. FidelityLoop makes both policies and their evaluators testable against physical deadline and resource outcomes. We release paired traces from 197 physical runs and CPU analysis tools under an explicit resource and synthetic-API cost contract.

1 INTRODUCTION

Elastic LLM serving releases idle instances to save resources, but capacity returns only after startup. Queues, running work, and lifecycle transitions then shape subsequent control decisions. A simulator used to evaluate such policies must capture when capacity is available, not just how long individual requests take or how many GPU seconds a run consumes.

FidelityLoop connects prediction, policy decisions, and physical lifecycle execution through a shared contract (Figure 1). It tests whether a replay model preserves the outcomes that matter for policy selection: deadline feasibility and the resources consumed by feasible executions. Our measurements reveal two ways aggregate accuracy can mislead this choice. Startup timing can cross a controller's cooldown boundary and change subsequent actions; resource errors can also cancel in the total while capacity is available at the wrong times. Across policies, arrivals, and serving contexts, controlled physical measurements connect these errors to deployment choices and costs. The framework makes this diagnostic process reusable.

The paper makes three contributions:

1. **A closed-loop evaluation framework.** FidelityLoop separates the policy adapter from lifecycle execution and aligns causal observations, request identities, lifecycle generations, resource trajectories, and deadline-aware cost ledgers. Its adapters connect rules and learned policies to vLLM and SGLang, and can substitute Vidur service predictions in replay. Aligned traces retain failed requests in deadline denominators.
2. **A lifecycle-fidelity diagnosis.** Held-out physical runs distinguish local-service, aggregate-resource, and trajectory fidelity. Event-level comparisons explain a startup-related divergence and show why aggregate resource error is insufficient for a closed-loop evaluator.
3. **A reproducible policy evaluation.** A deadline guard and frozen PPO checkpoints test feasibility and resource–latency trade-offs. Follow-up checkpoint, arrival, serving-stack, and model-size studies test the evaluator's transfer across contexts. Retrospective deployment choices quantify the consequences of false accepts and false rejects. We release paired prediction and execution traces with CPU analysis tools.

The release provides CPU interface checks, a fixed-event

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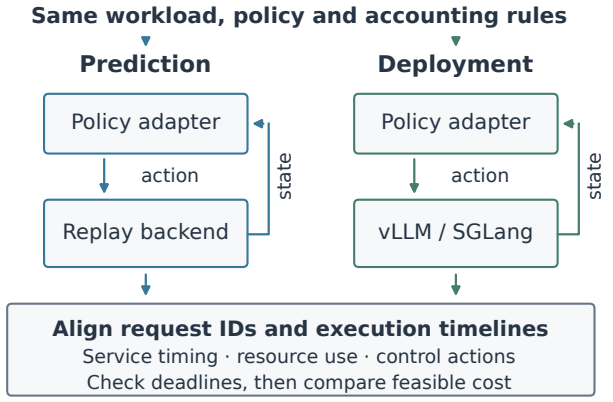


Figure 1. FidelityLoop runs the same policy contract in separate replay and physical control loops. Each adapter reads its own backend's causal state. Aligned records expose errors in both predictions and policy choices; physical outcomes are used for validation after execution.

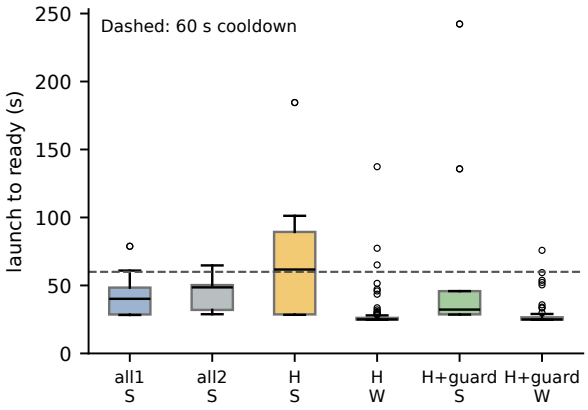


Figure 2. Observed startup times in the held-out runs. Labels use S for setup and W for window scale-out. The dashed line marks the 60-second controller cooldown; boxes summarize observed launch-to-ready times. Release and cleanup are separate ledger events. Collapsing these contexts into one endpoint can alter closed-loop actions.

cost verifier, and a paired trace corpus[1]; the supplement documents their reproduction scopes.

2 Background and Motivation

We consider mixed online and offline LLM workloads served by a limited GPU pool, where elastic scheduling aims to reduce resource use while meeting request deadlines. Evaluating every candidate policy on physical hardware is costly, motivating a framework for replay-based policy development and training followed by physical validation. FidelityLoop connects these stages through a shared policy interface and consistent lifecycle and deadline accounting, so that savings predicted in replay can be checked against deployment outcomes.

Our exploratory deployment revealed why this connection matters: the calibrated replay reduced prediction error for matched local requests while increasing error in occupied GPU time under dynamic scheduling. These preliminary observations motivate the preregistered study; they are not additional held-out repeats.

The lifecycle diagnosis separated the old 69.219 s startup proxy from the earlier window launch-to-ready (25.19 s) and setup (46.80 s) contexts. When the proxy crossed the 60 s cooldown boundary it changed a target decision, motivating the locked sequence of calibration, lifecycle checks, and held-out feasibility tests.

Figure 2 summarizes this lifecycle boundary; the held-out selection procedure is fixed separately in Section 5.

[1] https://anonymous.4open.science/r/FidelityLoop-144B/

3 Related Work

Vidur and Charon model LLM execution to guide configuration search; Clockwork targets predictable request timing (Agrawal et al., 2024a; Yang et al., 2026; Gujarati et al., 2020). FidelityLoop's evaluation unit is an elastic policy rollout: its deadline classification, resource and control trajectories, and resulting deployment choice. Its shared policy interface and identity ledger align these outcomes between replay and physical execution.

Cold-start characterization decomposes initialization latency, while ServerlessLLM reduces startup overhead (Kabakibo et al., 2026; Fu et al., 2024). We study how errors in that latency change subsequent capacity decisions and deadline outcomes; the replay endpoint contrast holds the engine and loading path fixed. vLLM, SGLang, and Sarathi-Serve optimize request execution (Kwon et al., 2023; Zheng et al., 2024; Agrawal et al., 2024b); we validate elastic policies across execution backends and replay service models.

Provisioning and scheduling work connects workload estimates to SLOs (Crankshaw et al., 2020; Zhang et al., 2019; 2023). We evaluate whether replay models preserve policy feasibility and resource outcomes after deployment. The guard serves as a rule-based case study; frozen PPO checkpoints extend the evaluation to learned policies.

Policy design provides a complementary layer. Llumnix dynamically reschedules LLM requests (Sun et al., 2024); AAPA predicts scaling needs for serverless workloads (Zhang et al., 2025b); KIS-S learns GPU-inference autoscaling with PPO (Zhang et al., 2025a). We use PPO as a frozen policy family rather than proposing a new RL algo-

rithm; PPO is the standard clipped policy-gradient method (Schulman et al., 2017). Our question is whether a learned checkpoint selected in replay remains feasible after the lifecycle backend and deadline ledger are applied.

4 PROBLEM AND METRICS

The system has two accelerators, online arrivals with a 60 s deadline, and 120 offline jobs per 2,100 s observation window with a 300 s deadline. Online requests run locally or use a delay-only synthetic API sink; offline jobs run locally. Arrivals span 1,800 s; execution and accounting continue for another 300 s to cover the remaining deadline horizon. The primary window cost, in scenario USD, is

$$C = .001G + .002S + .001D + 10^{-6}I + 3\times10^{-6}B + .01M, \quad (1)$$

where G is occupied GPU seconds, S, D are starts and shutdowns, I, B are synthetic API input and output budgets, and M counts offline deadline misses, including unfinished requests whose deadlines have passed. Setup and cleanup are reported separately and are not folded into the primary model gate.

Completed means a recorded completion by the observation cutoff; it may be *on time* (finish time $\leq$ deadline) or *late* (finish time $>$ deadline). An unfinished request has no completed terminal by the cutoff. Unfinished or censored work is retained in the arrival denominator. The capacity gate G_F requires all2 (two GPUs throughout) to complete every online and offline request on time. P1 requires 120/120 offline requests on time, at least 99% online requests on time, and no unfinished request; P1+ additionally requires every online request on time. P2 is a ratio-of-means saving against all1 (one GPU throughout) computed within each window and then averaged equally across the three windows.

5 WORKLOAD CONSTRUCTION

Online arrivals in the main study come from the successful-request subset of BurstGPT, a public trace of production LLM requests (Wang et al., 2024). We read all 1,404,294 source rows and retain source hashes and row identities. The trace has no session identifier, so the split separates arrival times rather than sessions. Locally routed requests execute actual GPU inference with fixed-corpus prompts and registered token budgets. The experiment reproduces arrival-time patterns, not the original users' prompts or semantic tasks.

The three held-out windows are selected once from the forward pool after the old windows plus a 30-minute protection band. A 60 s morphology filter defines the steady, recovery, and burst-offline classes (even, declining, and bursty online arrivals); a fixed SHA ordering takes the first eligible candidate in steady–recovery–burst order, subject to the registered separation constraint. Online rows are thinned with the inherited C2 integer threshold, while the 120 offline jobs are constructed at fixed 15 s spacing and 256-input/256-output token budgets. The two salts and all thresholds are fixed in `s4_generation_lock.json`; generated inputs are locked before replay and never replaced to improve a result.

Table 1. Selected window arrival shape at 60 s bins. Each cell is raw/derived; CV and peak-to-mean use the full 1,800 s window, including empty bins.

Window	arrivals	CV	peak/mean	lag-1
steady	5461 / 1444	0.378 / 0.412	1.566 / 1.641	0.772 / 0.797
recovery	1141 / 334	1.444 / 1.452	3.550 / 3.862	0.906 / 0.892
burst-offline	663 / 177	1.951 / 1.935	6.652 / 6.780	0.810 / 0.754

Table 1 reports the arrival shape at the registered 60 s scale. The 1 s diagnostics are retained in the manifest but do not define the class. The values describe the selected derived windows, not the full source population.

These derived windows represent a controlled lifecycle workload domain, not the full production population or original user tasks.

The checkpoint-deployment study (E2) uses two separately generated and locked windows from the same construction pipeline, with no row overlap with the three held-out windows; the campaigns remain distinct and are never pooled. The checkpoint-screening study (E3), second-engine study (E4), and 14B port (E6) reuse these two windows. The recovery window contains 358 derived online arrivals and 120 fixed offline arrivals (478 total); the steady window contains 944 online and 120 offline arrivals (1,064 total). Offline arrivals are spaced every 15 s, with 256 input and output tokens; the online rows retain their locked arrival timestamps. The window identities and input hashes are bound in the E2 lock.

A separate external-validity study uses Azure conversation arrival traces associated with Splitwise (Patel et al., 2024); its workload construction and load-boundary results are reported in the supplement.

6 EXPERIMENTAL SETUP

The preregistered held-out study used two NVIDIA A100 80GB PCIe GPUs at a university computing facility, with one independent tensor-parallelism-1 instance per GPU and Qwen2.5-7B-Instruct. The serving stack was vLLM 0.9.0 (Kwon et al., 2023) with PyTorch 2.7.0+cu126, Transformers 4.51.1, tokenizers 0.21.1, and NumPy 1.26.4. Instances used FP16, maximum model length 4096, maximum

batched tokens 16,384, maximum sequences 1,024, four local request slots, chunked prefill enabled, prefix caching disabled, eager execution enabled, and GPU memory utilization 0.8. Readiness required the per-process health probe and the dual-ready barrier; cleanup required verified process release on both devices. These values are bound by each run's environment snapshot and GPU identity record.

The held-out study and E2 both use two A100 GPUs. Before held-out execution, the deployment-adapted model (E) had its lifecycle parameters calibrated and frozen. The historical V–E comparison spans different hardware environments (A800 and A100), so it does not isolate the effect of calibration.

E2 uses the same Qwen2.5-7B-Instruct stack under a separate runtime lock. Its 32 cells comprise one all2 capacity anchor and five frozen PPO arms per window, with three repeats per comparison arm. Checkpoint and tensor identities are bound into the results. An earlier 38-run campaign, termed Bridge, evaluated rule-based, guarded, and PPO policies under a separate runtime lock; its results are retained as supplementary evidence and are not pooled with E2. The follow-up screening study (E3) uses six previously undeployed checkpoints on the same two windows. Each window runs on a separate dual-A100 allocation with the registered serving stack: one all2 anchor and two repeats per arm, 26 cells in total. E3 is analyzed separately from E2 and Bridge.

E4 runs SGLang 0.4.8 with PyTorch 2.7.1 and Transformers 4.52.3 on two A100 80GB PCIe GPUs. It reuses the Qwen weights, two windows, policy parameters, and accounting contract: six policy arms run twice per window, plus two all2 anchors (26 cells). It evaluates transfer of the complete serving stack; versioned configuration and token-input compatibility details appear in the supplement.

E5 keeps the SGLang/7B stack and freezes C, C30, and C_{sg} before three additional windows (27 cells, including 24 policy cells). E6 ports the contract to Qwen2.5-14B-Instruct/vLLM on the two locked windows (12 cells), freezing two predictors that differ only in startup; neither extension retrains PPO or searches policy parameters.

7 METHOD

Predictive replay and physical execution share a policy contract. Adapters read observable queues and device states and return capacity targets; the guard may also select dispatch priority. The backend handles readiness, draining, and verified release, while request identities align predictions, decisions, lifecycle states, resources, and costs. A separate CPU verifier recomputes feasibility and cost from raw records. Engine adapters translate launch, readiness, inference, and release operations into this contract, allowing a serving backend to change without redefining policy observations or evaluation metrics. By default, ready local slots serve online requests before offline jobs. Online requests still waiting 10 s after arrival can enter the synthetic API sink, which admits up to eight concurrent requests and completes each on a 5 s timer. The guard can prioritize at-risk offline jobs; PPO changes capacity while retaining the default dispatch and API rules.

Table 2. Frozen replay parameters in seconds, rounded to three decimals. Listed models share a 69.363 s setup proxy outside the primary window ledger.

Model	Service	Startup	Shutdown
O	equal-share	30.000	7.059
C	measured lookup	69.219	7.059
C30	same lookup	30.000	7.059
E	same lookup	33.497	2.224
V	same lookup	25.196	2.325

The policy and backend adapters have separate interfaces; the supplement reports their implementation footprint. CPU checks preserve historical request/event ledgers, and causal replay reproduces all 29,580 logged decisions from 16 same-runtime runs. A Vidur service backend (Agrawal et al., 2024a) supplies native scheduling and token execution while FidelityLoop retains admission, lifecycle transitions, and the frozen guard. A Qwen2.5-7B/A100 service screen precedes the retrospective policy comparison in Section 8.1; the supplement provides profiles and parameters.

7.1 Calibrated models

The replay models share an executor but differ in service and lifecycle timing (Table 2). O uses an equal-share service model and an assumed 30 s startup. C uses a measured service lookup and a historical 69.219 s startup proxy; C30 restores the 30 s assumption. E combines the lookup with deployment-adapted lifecycle parameters; V retains a bounded refit from the earlier environment. O–C30 isolates the service model, whereas C–C30 isolates startup; O–C changes both.

O assumes 1024 input and 128 output tokens/s, shared equally among concurrent requests. The calibrated lookup uses median measured wall times per input/output/concurrency cell, interpolating in input length and concurrency (one to four requests per instance). Its values are shared by C, C30, E, and V. The supplement's Replay model specification section details measurement sources, interpolation, and lifecycle fitting. Model bytes, tokenizer, workload identities, and the candidate grid are locked before held-out execution.

For E4, C_{sg} replaces C's startup proxy with the median

of 20 SGLang scale-outs measured while the other GPU serves a fixed background load: 27.568 s. It retains C's service lookup and shutdown parameter. C, C30, and C_{sg} predictions are frozen before the policy runs; the guard retains its separate safety estimates.

E5 reuses these three models without recalibration. In E6, the historical-proxy predictor S14 and locally calibrated L14 share freshly measured 14B service, setup, shutdown, and guard parameters; only startup differs (69.219 s versus 46.067 s). Both are frozen before physical execution.

For each model prediction and physical repeat of a window, the dimensionless lifecycle loss is

$$L = \frac{0.5|\Delta G_{\text{occupied}}| + 0.25|\Delta G_{\text{active}}|}{2T} + \frac{0.25|\Delta Q|}{TN}, \quad (2)$$

Here G_{occupied} sums per-GPU time from launch issuance to verified release, while G_{active} sums time in the active state, including idle time; both are measured in GPU seconds. The difference Δ is prediction minus observation, $T = 2100$ s for formal windows, and N counts *all* online and offline arrivals. Q is the time integral of the released-but-undispatched queue, in request seconds. Unfinished requests remain in N and contribute queue time while queued up to cutoff. Two devices give the $2T$ normalization. We first average losses over repeats within each window, then weight windows equally; we do not pool requests or restrict N to matched-local support. L measures prediction error, separately from scenario cost C; the shorter development horizon is stated with that experiment.

7.2 PPO policy contract

The learned policy is trained only in the CPU replay environment, with C or C30 as the training environment; the physical serving loop is used only after a checkpoint is frozen. At each one-second decision boundary, its 30-dimensional causal observation contains normalized time and target state, queue counts and ages/slack, device lifecycle states and ages, running-job summaries, and the API queue. The capacity action proposes a target of 0, 1, or 2 GPUs; the registered 60 s cooldown filters that proposal before execution, while the offline action head is fixed to its sole legal value. The reward is the negative incremental scenario-ledger cost, so training trades occupied resources, lifecycle events, API use, and deadline misses under the declared contract.

Training uses the three development arrival regimes and paired seeds from the locked CPU PPO contract. The five E2 checkpoints are selected mechanically with C30 replay, guard disabled, P1+ required in every validation window, and sorting by equal-window mean cost, checkpoint index, then seed. We label the frozen checkpoints C-1/C-2 and C30-1/C30-2/C30-3 by training environment, numbering them by seed and then episode, independently of physical outcomes. Training settings, data partitions, and checkpoint coordinates are detailed in the supplement's Frozen PPO training and selection contract section.

7.3 Controller and guard

H uses up-threshold 8, down-threshold 0, and a 60 s cooldown. At each one-second tick, H reads queue and running counts plus its target/cooldown state, not the predictor's startup parameter. The guard's lifecycle-delay estimates use separately frozen safety parameters, distinct from the C/E simulator endpoints. The guard observes only released queues, device states, and running-job ages; future arrivals and true completion or readiness times are unavailable. Let P_t be the pending offline request IDs and b_t H's current target after its decision. The projection $F_j(k, r)$ estimates request j's finish with k devices and reservation setting r, using frozen lifecycle delays, running-job residuals, and concurrency-four service estimates. It schedules pending jobs by deadline then request ID over four slots per device. Online backlog precedes offline work unless reservation ($r = 1$) assigns one slot to offline jobs; without online backlog, offline work can use all slots. The supplement's "Guard implementation map" specifies device selection and overdue-estimate fallbacks. These estimates are not proved completion-time upper bounds.

For the full guard ($r = 1$), margin $m = 5$ s and previous risk-ID set A_{t-1}, the update is

$$\begin{aligned}
R_t &= \{j \in P_t : F_j(\max(1, b_t), 1) + m + 1 \geq d_j\},\\
A_t &= (A_{t-1} \cap P_t) \cup R_t,\\
K_t &= \{k \in \{1, 2\} : \forall j \in P_t,\ F_j(k, 1) + m \leq d_j\},\\
u_t &= \begin{cases} b_t, & A_t = \emptyset,\\ \max(b_t, \min K_t), & |A_t| > 0,\ |K_t| > 0,\\ 2, & |A_t| > 0,\ K_t = \emptyset. \end{cases}
\end{aligned} \quad (3)$$

Here d_j is the deadline. Triggering includes a one-second lookahead and uses $\geq$; candidate feasibility uses $\leq$ without that extra second. Empty K_t with active risk sets an `infeasible` flag. A latched ID is removed only when it leaves the pending queue, not when a later estimate becomes safe. The target never falls below *this tick's* H target. A guard raise can bypass H's cooldown and writes u_t and the current time back to H's target and last-change state; it does not guarantee global deadline feasibility.

While A_t is nonempty and both queues have work, dispatch reserves at most one offline slot. An already-running offline job leaves other slots online-first. Otherwise, for head-request slacks $s_x = d_x - t - \widehat{p}_x$, online wins only if $s_{\text{on}} \leq m$ and $s_{\text{on}} \leq s_{\text{off}}$. Without online backlog, offline dispatch is unrestricted. No running request is preempted; later scale-out must wait for readiness.

The registered development search covered 26 controller configurations, including guard variants. Selection preceded held-out input access and returned the preregistered H+guard configuration; its duplicate arm was merged into the four-arm formal matrix to avoid pseudo-replication.

7.4 Evidence protocol

The held-out study evaluates all1, all2, H, and H+guard on steady, recovery, and burst-offline windows, with three repeats per arm (36 formal runs). CPU verification uses the original acceptance code and lock-bound sources. Two operator-aborted attempts remain in history but not formal statistics. Metrics use run and window units, not independent-request sampling. Bridge's terminal counts, lifecycle occupation, and costs are recomputed from raw events; sealed summaries provide consistency checks. A technically valid negative result remains in the matrix.

The fixed gate order starts with full-capacity feasibility (G_F), then tests occupied-resource error (M1), deployment-adapted lifecycle loss (M1b), and matched-request service fidelity (M2). Target agreement (M3) is diagnostic. P1/P1+ and P2 then assess policy feasibility and cost, retaining H's failures as control evidence. This order separates model validity from the appeal of a policy's predicted savings.

8 RESULTS

We first establish the prediction-level mismatch that motivates the framework. We then test replay feasibility in policy deployments, across new checkpoints, and across serving contexts. The guard case study connects these validation results to the resource–latency trade-off.

8.1 Cross-level prediction fidelity

On fixed H, E's occupied MAE is 67.94 GPU seconds versus 543.51 for O (87.50% lower). C30's improvement over C implicates startup. V reduces equal-window L by 34.24% relative to E, but E uses deployment-adapted endpoints while V retains the earlier environment snapshot. M1b is therefore not a clean same-environment causal comparison.

For local service fidelity, the fixed all1/all2 tests give a maximum window MAE of 0.064 s for C, C30, E, and V, while O reaches 3.016 s. On dynamic arms, the minimum common support across model/route projections is 122 requests and the maximum common-support MAE is 0.129 s for C, 0.112 s for C30, 0.110 s for E, and 0.112 s for V; O reaches 4.510 s. Common support is the intersection of request IDs completed locally in the compared predictions and physical runs; coverage and routing discrepancies are reported in the artifact.

Table 3 reports window-wise occupied-resource errors for fixed-H replay, distinct from the route-conditioned matched-ID checks.

Table 3. Occupied-resource MAE (GPU seconds) by window and model. L is the equal-window lifecycle loss defined above.

Model	steady	recovery	burst	Occupied MAE	L
O	1151.97	244.08	234.48	543.51	0.09403
C	964.14	864.20	847.60	891.98	0.12162
C30	67.18	78.30	83.08	76.19	0.01353
E	66.34	66.95	70.55	67.94	0.01353
V	43.51	65.74	69.34	59.53	0.00890

The C-versus-O comparison reproduces the exploratory divergence: maximum matched-ID service MAE falls from 4.510 to 0.129 s (about 35-fold), while equal-window occupied-resource MAE rises from 543.51 to 891.98 GPU seconds (64.1%). The aggregate deterioration does not occur in every window. O–C changes both service and startup; the C–C30 contrast isolates startup, lowering occupied MAE to 76.19 GPU seconds.

Retrospective analysis of the frozen trajectories explains this endpoint effect and its limits (supplement). In all three windows, C/C30 targets first diverge 60 s after a common instance start: C remains unready with higher queue/running pressure; C30 is ready. This simulator contrast is not a physical intervention on startup or cooldown. Six burst/recovery physical repeats first disagree in target with predictions at 3 s, before any window restart. Conversely, steady repeats 1 and 3 match C30 targets throughout despite state, routing, and occupation errors.

Aggregate agreement can mask temporal misalignment. In steady, E and C30 have nearly equal total occupation MAE, but E's mean time-integrated absolute error in occupied GPU count is 1390.504 versus 114.324 GPU seconds (Table 4). Positive and negative count errors cancel in the total; their timing still matters to control. Together with the event-level counterexamples, this shows why neither startup correction nor target agreement alone establishes fidelity. The supplement retains all 36 prediction/physical pairs and 12 model/window means; this diagnosis neither changes M1 nor reselects a model. The later Bridge cells are a separate policy-contract study.

Replacing the service lookup with Vidur's profiled model tests the same validation contract with an external predictor. Twelve retrospective CPU rollouts retain E's lifecycle parameters and the original guard, and reuse all 36 physical runs. They reproduce H's recovery misses (116/120 offline jobs on time versus 113–116 physically) and the guard's 120/120. For H and H+guard in steady, median occupied-resource absolute percentage error (APE) is 2.47%, yet

Table 4. Steady-window fidelity: means over the same three physical repeats. Resource errors are GPU seconds; target disagreement is a percentage of decision ticks.

Model	Total error	Trajectory error	Target diff. (%)
C30	67.18	114.32	1.03
E	66.34	1390.50	66.73

Table 5. E2 policy execution on two A100 GPUs. P1+ requires every online and offline request to meet its deadline. Costs are registered-window scenario USD under the declared ledger. R/S report P1+ passes/runs in recovery/steady. The last column is cost reduction relative to all2; † marks a failed arm and is descriptive only.

Arm	R	S	Cost R	Cost S	Cost ↓ (%)
all2	1/1	1/1	4.200	4.200	0
C-1	3/3	3/3	2.357	2.449	42.8
C30-1	3/3	3/3	2.073	2.084	50.5
C30-2	3/3	3/3	2.051	2.103	50.5
C30-3	0/3	0/3	1.899	2.090	52.5†
C-2	0/3	0/3	2.730	2.950	32.4†

target agreement is 33.0–35.5%; guard recovery has 14.74% occupied-resource APE. Thus the external backend preserves the key deadline contrast while revealing residual timing errors (supplement).

These rule-policy tests establish the need to check both outcomes and trajectories. We next let frozen learned checkpoints drive the same physical lifecycle and compare their predicted and observed feasibility and cost.

8.2 Policy execution and screening: feasibility is part of the result

E2 tests whether checkpoints selected in replay remain feasible in physical execution. In its re-provisioned runtime, 196 of 218 window restarts that reached readiness across the 30 PPO runs took over 60 s. Each window begins with an all2 capacity anchor; the five PPO arms then run in three repeats. All 32 cells are technically valid, and the package contains 36 attempts, including four preserved technical failures. C30-2 is the preregistered reference checkpoint. The table reports comparison means and one all2 run per window.

Three PPO arms pass all six physical repeats: C-1, C30-1, and C30-2; both all2 anchors also pass. The three feasible PPO arms have scenario costs 42.8–50.5% below all2, but this comparison is conditional on the declared ledger and synthetic API sink. C-2 is a decisive negative case: it averages only 4–7 on-time offline requests out of 120 and leaves

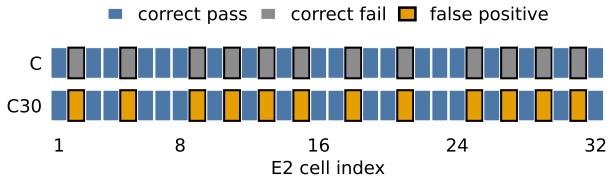


Figure 3. Replay-to-deployment feasibility for all 32 E2 cells. Blue and gray tiles are correct pass/fail directions; orange tiles are C30 false positives. Black outlines mark cells that physically fail P1+.

about 80 unfinished, despite lower cost. C30-3 is a sharper boundary: it completes the offline stream eventually, but roughly 11 of 120 offline requests miss their deadline per repeat, so its lower cost is not feasible saving. Online requests are on time in every cell; the E2 failures are therefore isolated to the offline deadline path rather than hidden by an online regression.

Predictions were frozen before physical execution. Table 6 compares each cell with its corresponding replay: C correctly predicts all 20 passes and 12 failures, whereas C30 predicts all cells feasible and misses both negative checkpoints. For each replay model, these 32 comparisons reuse 12 policy/window predictions; the physical repeats are not independent predictions. The two replay models have window-cost MAE of 0.0804 and 0.4657 scenario USD, respectively. C-2 exposes a startup–cooldown interaction: the 60 s cooldown begins at the capacity-target change. In C30 replay, readiness precedes cooldown expiry, leaving time to serve queued work; in C replay and most physical restarts, readiness follows expiry, allowing almost immediate scale-down. Across the six physical C-2 runs, 72 of 73 restarts taking over 60 s enter draining within two seconds of readiness (see the supplement's Retrospective E2 lifecycle diagnosis section).

The earlier 38-cell Bridge campaign contains rule, guard, and PPO arms on a separate runtime. C also has higher feasibility agreement and lower cost MAE there (Table 6); the campaigns are not pooled. This repeated agreement motivates testing C as a screening rule on new checkpoints.

Both training environments yield passing and failing E2 checkpoints. E3 tests C on six previously undeployed checkpoints in a new execution context. From 90 eligible candidates, C predicts 67 feasible on both known windows and rejects 23. A fixed ordering selects three distinct trajectories from each group; C30 does not enter selection. All six pass every physical run (two windows, two repeats), as do both all2 anchors. C therefore falsely rejects three feasible candidates, while C30 accepts all six. Across the 24 policy cells, C30 also has lower cost MAE (Table 6), reversing the E2 ordering. These selected cases show that earlier agreement does not establish reliable screening of new checkpoints

Table 6. Replay-to-deployment agreement and window-cost MAE for the historical campaigns. Bridge and E2 include all technically valid cells, including anchors and deadline failures; E3 includes its 24 policy cells. Cost MAE is in scenario USD.

Campaign	Replay	Agreement	Cost MAE
Bridge	C	36/38	0.0969
Bridge	C30	21/38	0.6998
E2	C	32/32	0.0804
E2	C30	20/32	0.4657
E3	C	18/24	0.3112
E3	C30	24/24	0.0163

Table 7. E4: frozen predictions versus 24 SGLang policy runs. False accepts are out of two infeasible runs; false rejects out of 22 feasible runs. Cost MAE includes the declared deadline penalty, in scenario USD.

Replay	Agreement	False accepts	False rejects	Cost MAE
C	16/24	0/2	8/22	0.5420
C30	24/24	0/2	0/22	0.0468
C_{sg}	24/24	0/2	0/22	0.0427

and execution conditions; they do not estimate population screening accuracy. The supplement records candidate selection, predictions, and the separate window allocations.

8.3 Transfer across serving contexts

E4 tests whether the evaluation contract can carry unchanged policies to SGLang and still distinguish feasible from infeasible executions. The six arms are H, H+guard, E2's C-1/C-2, and two E3 checkpoints selected by lowest historical mean cost within C's accepted and rejected groups. All 26 cells are technically valid. Both all2 anchors and all guarded and PPO runs satisfy P1+; H meets every deadline in recovery but leaves one offline job unfinished in each steady repeat.

The historical C proxy falsely rejects eight feasible runs, whereas C30 and C_{sg} match all 24 outcomes, including H's two failures (Table 7). Measuring startup locally slightly lowers cost error relative to C30 but leaves feasibility judgments unchanged. Each model contributes 12 policy/window predictions, each compared with two physical repeats. Full results appear in the supplement.

E5 tests whether E4's calibration remains useful on additional arrivals, without refitting. On three new constructed windows, all 24 policy runs are feasible: C, C30, and C_{sg} agree on 14/24, 22/24, and 24/24, respectively. C_{sg} thus avoids the false rejections here; this all-feasible cohort does not test detection of infeasible policies.

Table 8. Formal held-out results. Each arm has three repeats per window; H has 15 offline deadline misses across its three recovery repeats.

Arm	tech.	offline	online
all2	9/9	9/9 (120/120)	9/9
all1	9/9	9/9 (120/120)	9/9
H	9/9	6/9	9/9
H+guard	9/9	9/9 (120/120)	9/9

E6 tests remeasured lifecycle timing in Qwen2.5-14B-Instruct/vLLM. H fails all four runs; H+guard passes all four. Replacing S14's historical startup with the measured value in L14 raises agreement from 4/8 to 8/8 and lowers cost MAE from 0.4105 to 0.0668 scenario USD. A retrospective 30 s startup contrast, holding all other 14B fields fixed, reaches 6/8: it still falsely accepts H in steady. Each model supplies four policy/window predictions reused across two repeats. Event replay shows how longer startup changes control: S14 retains queue pressure through cooldown expiry and raises the target to two GPUs, altering later dispatch and deadline outcomes. The supplement traces these feedback paths and distinguishes them from physical-state mismatches that precede restart.

These errors also affect deployment choices. Retrospectively requiring predicted P1+ in every window, then minimizing mean predicted cost, selects E2's C-2 under C30; it fails all six physical runs. In E4, C selects a feasible policy costing 58.95% more than the lowest-cost feasible policy in the tested pool, which C30 and C_{sg} select. Across a 3×3 GPU/API price grid, E2's failing choice persists and E4's gap ranges from 16.56% to 80.17%. The supplement reports all choices and ties under the scenario ledger. The interfaces transfer across serving contexts; physical outcomes validate both predictions and their consequences for policy selection.

8.4 Control case study: feasibility and cost

The guard provides a rule-based control case alongside the learned policies. Table 8 summarizes the original held-out study. All2 and all1 satisfy the registered completion target. H is technically valid but fails the P1 gate in all three recovery repeats (116/120, 113/120, and 116/120 offline on time). In that held-out study, the guard repairs this failure without changing the workload or the locked model.

H+guard's window cost is lower than all1 by 21.96%, 48.19%, and 58.08% in steady, recovery, and burst-offline, respectively. Their equal-window mean is 42.74%, above the preregistered 25% P2 line, although steady falls below that stretch target. Including setup and cleanup gives a separate mean saving of 38.86%.

The primary ledger covers the same 2,100 s interval; the

Table 9. Guard activity by recovery repeat. Ticks are window seconds; slack is the minimum pending-offline time to deadline (seconds), before subtracting service time.

Repeat	risk ticks	target raises	tick range	slack range
1	30	1	1982–2011	29–58
2	0	0	–	–
3	29	1	2027–2055	30–58

deployment ledger also charges setup and cleanup. The supplementary table retains both views, with unchanged prices and event streams.

Before execution, E's locked recovery prediction identifies four unfinished offline requests; the three physical H repeats leave four, seven, and four unfinished, for 15 total misses. This prediction is interpreted alongside, not substituted for, the observed completion evidence.

8.5 Latency–cost trade-off

Resource savings consume online latency margin: per-repeat p99 is about 1.6–2.6 s for always-on references and 15–19 s for both dynamic arms (supplementary tail table). Requests routed to the synthetic API sink wait at least 10 s before admission; the guard protects offline deadlines without removing this delay. All observed p99 values remain below the registered 60 s online deadline, making the saving an explicit SLO-margin-for-cost trade-off.

8.6 Guard activation and mechanism

The guard checks deadline risk using the visible queue, running-job ages, lifecycle delay, and concurrency-four service estimate. Equation 3 retains risk IDs until dispatch or removal from the pending queue and distinguishes risk from a target raise. In the held-out guard study, the logs contain 59 risk-active ticks, all in recovery: the guard raised the target above H twice, with the remaining risk ticks maintaining the already-raised target. Its steady and burst-offline runs have no risk-active ticks. In the 14B study, active guard runs in recovery and steady restart capacity for tail queues below H's scale-out threshold (supplement). Table 9 reports the held-out recovery counts and slack ranges.

8.7 Price sensitivity

Repricing the same H+guard and all1 event streams over 27 window–GPU–API price cells yields one negative saving: steady at half-price GPUs and twice-price API use (-39.59%). Its break-even API multiplier is 1.004. Recovery and burst remain positive throughout the tested grid; the supplement retains the full table. These are scenario sensitivities, not provider billing forecasts.

External checks cover queue-target (B-HPA), Holt-predictive (B-PRED), and queueing-based (B-MMC) baselines, Llama, and Azure arrivals (supplement). Each baseline meets all deadlines in nine runs; H+guard costs less than B-HPA on two known Qwen windows, with higher API use and queue exposure. Retrospective replay of 24 Azure runs gives archived C, C30, E, and V identical 21/24 agreement, but C falsely accepts three runs while the others reject three feasible runs. H's physical repeats have mixed outcomes in both windows. Error direction therefore matters alongside aggregate agreement.

9 DISCUSSION

FidelityLoop evaluates whether policies meet deadlines and what resources feasible executions consume. C30 reduces fixed-H resource error, whereas C matches feasibility better in Bridge and E2. Tests on new checkpoints, SGLang, and a 14B vLLM port show why predictions must be revalidated in each deployment context.

9.1 Validating predictions across levels

Lifecycle validation should follow the controller's clock. For controllers that block scale-down until a target-change cooldown expires, readiness after expiry may permit immediate scale-down even when replay predicts a service interval protected from scale-down. E2 exhibits this interaction. Comparing deployment startup timing with controller boundaries is therefore a useful diagnostic; setup and window scale-out should be measured separately. State and action trajectories also need inspection because aggregate resource error can hide temporal misalignment.

The guard shows how a controller can act on deadline risk from observed queues and frozen safety estimates. It repairs H's held-out recovery failure, but H does not read the simulator's startup parameters: the endpoint correction and the guard benefit have distinct causal roles.

9.2 Resource and latency trade-offs

Elastic policies trade online latency margin for lower occupied capacity. The same-runtime comparison with B-HPA quantifies this trade-off in API share, lifecycle-event counts, and queue exposure. Price sensitivity and the Llama and Azure tests identify economic and workload boundaries; meeting one declared SLO does not establish suitability for tighter latency targets or heavier loads.

The supplementary CPU microbenchmark times `adapter.decide` on reconstructed states outside live serving. It measures decision-call latency, not full-controller or end-to-end overhead.

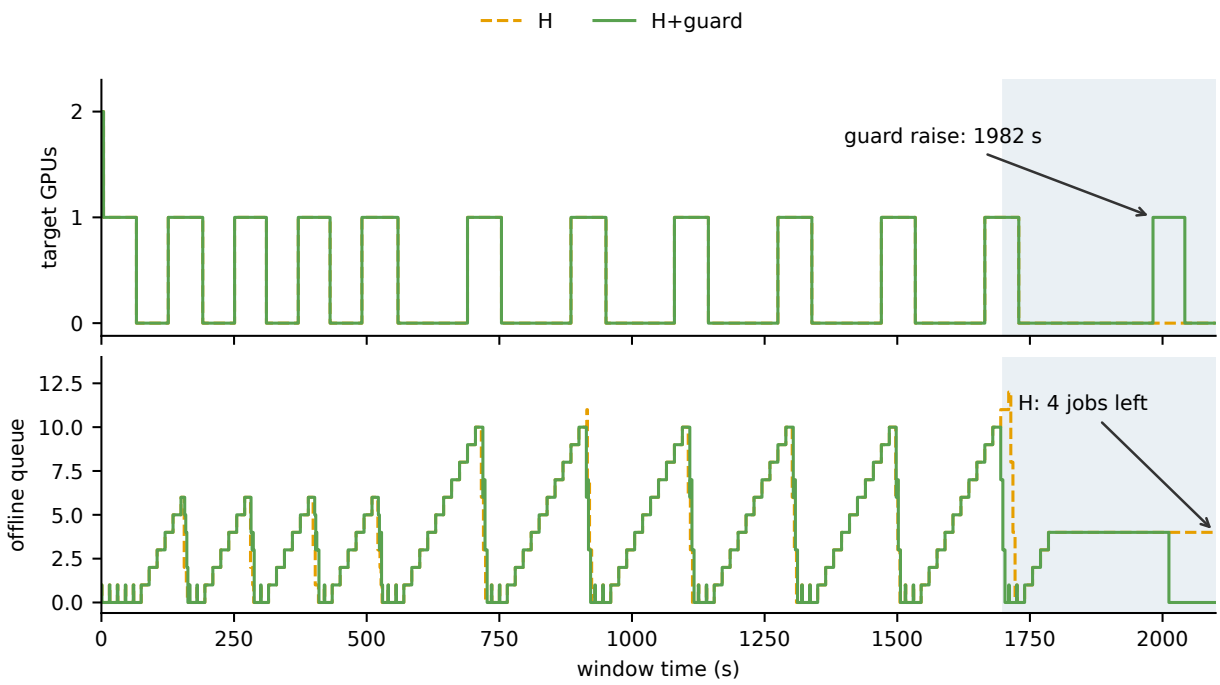


Figure 4. Recovery target and offline queue in repeat 1. The shaded final interval contains the guard raise at 1982 s; H leaves four offline jobs queued, while the guard clears them. Across all three H repeats, 15 jobs miss their deadlines.

Practical validation. Measure lifecycle timing for the deployment, then freeze predictions before physical execution. Check service timing, resource use, and control trajectories separately; report false accepts and false rejects, retaining each physical repeat. Condition savings on meeting deadlines. When arrivals or serving context change, repeat these checks under the same accounting contract.

10 Scope and Limitations

The synthetic API implements a five-second timer after routing wait, without answer generation or quality evaluation. Scenario costs thus measure the declared resource contract rather than equal-quality commercial substitution. GPU release denotes process and resource release, not physical power-off, energy consumption, or provider billing.

The studies cover two-GPU deployments, Qwen 7B/14B and Llama 8B, vLLM and SGLang, and selected trace-derived windows. Repeats measure execution robustness rather than independent workload coverage. Missing session identities, constructed offline pressure, and nonrandom checkpoint selection limit population-level inference; E3, E4, and E6 reuse known windows. E3 and E5 are all-feasible cohorts, while E4 and E6 contain two and four infeasible policy runs. The evidence does not establish large-scale heterogeneous deployment or a general ranking of replay models, PPO training environments, or PPO versus the guard; other model families and quantized inference remain untested.

Runtime identities remain separate: deployment drift confounds V–E, E3 uses one allocation per window, and E4 changes the engine and dependencies together. E6 isolates startup within the 14B predictor, not the causal effect of model size, engine, or hardware. The same-runtime baseline comparison includes dispatch differences, rather than isolating the guard. Guard repairs provide no general safety guarantee, as the Llama failure illustrates. The supplement retains technical-attempt history and detailed boundaries.

11 Conclusion

FidelityLoop provides a replayable contract for connecting elastic-serving policy decisions to lifecycle execution and resource/cost accounting. Service-time fit, aggregate occupation, and control trajectories reveal different errors; the framework validates them together.

The guard and frozen PPO checkpoints demonstrate how this contract evaluates deadline protection alongside resource and cost trade-offs. Tests on new checkpoints, arrivals, and serving contexts show why the evaluator itself needs validation: earlier agreement can give way to false rejections or false accepts. FidelityLoop makes both the policy and its evaluator testable against physical deadline and resource outcomes.

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